Binomial Distribution

ANSWER KEY

Section 1: Binomial Criteria & Single Probabilities

1 Which of the following is NOT a required condition for a variable X to have a binomial distribution $B(n, p)$?

a. **The probability of success changes with each trial** ✓

b. There is a fixed number of trials n

c. Each trial results in one of only two outcomes (success or failure)

d. Trials are independent of one another

In a binomial distribution, the probability of success p must remain strictly constant across all trials.

2 If $X \sim B(6, 0.3)$, calculate $P(X = 3)$ correct to 4 decimal places.

a. 0.1852 ✓

b. 0.2541

c. 0.3087

d. 0.1176

$P(X = 3) = \binom{6}{3}(0.3)^3(0.7)^3 = 20 \times 0.027 \times 0.343 = 0.18522 \approx 0.1852.$

3 If $X \sim B(5, 0.4)$, find $P(X = 0)$.

a. 0.0778 ✓

b. 0.0102

c. 0.0625

d. 0.1000

$P(X = 0) = (1 - 0.4)^5 = (0.6)^5 = 0.07776 \approx 0.0778.$

Section 2: Cumulative Binomial Probabilities

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4 If $X \sim B(4, 0.5)$, find $P(X \geq 2)$.

a. $\frac{11}{16} = 0.6875$ ✓

b. $\frac{1}{2} = 0.5000$

c. $\frac{5}{16} = 0.3125$

d. $\frac{3}{8} = 0.3750$

$P(X < 2) = P(0) + P(1) = \frac{1}{16} + \frac{4}{16} = \frac{5}{16}$. Then $P(X \geq 2) = 1 - 5/16 = 11/16 = 0.6875$.

5 A manufacturer knows that 5% of lightbulbs are defective. In a random sample of 20 bulbs, what is the probability that at most 1 is defective?

a. 0.7358 ✓

b. 0.6824

c. 0.8245

d. 0.9128

$X \sim B(20, 0.05)$.
$P(X \leq 1) = P(0) + P(1) = (0.95)^{20} + 20(0.05)(0.95)^{19} = 0.358486 + 0.377354 = 0.73584 \approx 0.7358$
.

6 A fair coin is tossed 6 times. Find the probability of obtaining at least 5 heads.

a. $\frac{7}{64} \approx 0.1094$ ✓

b. $\frac{6}{64} \approx 0.0938$

c. $\frac{1}{64} \approx 0.0156$

d. $\frac{11}{64} \approx 0.1719$

$P(X \geq 5) = P(5) + P(6) = \binom{6}{5}(1/2)^6 + \binom{6}{6}(1/2)^6 = \frac{6+1}{64} = \frac{7}{64} \approx 0.1094$.

Section 3: Mean & Variance of the Binomial Distribution

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7 A random variable $X \sim B(48, 0.25)$. Find its mean and standard deviation.

a. **Mean** $= 12, \sigma = 3$ ✓

b. Mean $= 12, \sigma = 9$

c. Mean $= 16, \sigma = 4$

d. Mean $= 12, \sigma = \sqrt{3}$

Mean $\mu = np = 48(0.25) = 12$. Variance $\sigma^2 = np(1-p) = 48(0.25)(0.75) = 9$. Standard deviation $\sigma = \sqrt{9} = 3$.

8 For a binomial distribution $X \sim B(n, p)$, the mean is 6 and the variance is 3.6. Find n and p.

a. $n = 15, p = 0.4$ ✓

b. $n = 20, p = 0.3$

c. $n = 12, p = 0.5$

d. $n = 24, p = 0.25$

$\sigma^2 = \mu(1-p) \Rightarrow 3.6 = 6(1-p) \Rightarrow 1-p = 0.6 \Rightarrow p = 0.4$. Then $n = 6/0.4 = 15$.

9 What is the maximum possible variance of a binomial distribution with $n = 40$ trials?

a. **10** ✓

b. 20

c. 5

d. 40

Variance $= np(1-p)$. The quadratic $p(1-p)$ is maximized at $p = 0.5$, giving maximum variance $= 40(0.5)(0.5) = 10$.

Section 4: Contextual Modeling & Problem Solving

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10 A treatment is successful for 80% of patients. If 10 patients are treated, what is the most likely number of successful outcomes (the mode)?

a. **8** ✓

b. 7

c. 9

d. 10

The mean is $np = 10(0.8) = 8$. For integer $(n+1)p = 11(0.8) = 8.8$, the mode is the integer part $\lfloor 8.8 \rfloor = 8$.

11 How many times must a fair die be rolled so that the probability of getting at least one 6 is greater than 0.9 ?

a. **At least 13 rolls** ✓

b. At least 10 rolls

c. At least 6 rolls

d. At least 18 rolls

$P(X \geq 1) = 1 - (5/6)^n > 0.9 \Rightarrow (5/6)^n < 0.1 \Rightarrow n\ln(5/6) < \ln(0.1) \Rightarrow n > \frac{\ln(0.1)}{\ln(5/6)} = \frac{-2.3026}{-0.1823} \approx 12.63$. Least integer is 13.

12 In an exam, 70% of candidates pass. Out of 8 randomly chosen candidates, what is the probability that exactly 6 pass, given that at least 5 pass?

a. ≈ 0.380 ✓

b. ≈ 0.520

c. ≈ 0.296

d. ≈ 0.420

$P(6 \mid \geq 5) = \frac{P(X=6)}{P(X \geq 5)}$. Calculating with $B(8, 0.7)$:
$P(5) = 0.2541, P(6) = 0.2965, P(7) = 0.1977, P(8) = 0.0576$. Sum $= 0.8059$. Ratio $= 0.2965/0.8059 \approx 0.368$ (approx 0.38).

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